

Drone Detection System — Research Report

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Purpose: Military Defense Project — Drone Detection using Raspberry Pi 4

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1. What We're Building

A ground-based drone detection system that uses a camera and AI to spot hostile drones in real-time, tell them apart from birds and planes, and trigger a response — all running on a **Raspberry Pi 4 (4GB, v1.2)**.

The goal: **90%+ detection accuracy** with stable FPS (frames per second).

2. How Real Military Systems Do It (Reference Systems)

AeroEye (by Drone Defence UK)

- Uses HD cameras + AI to detect drones **over 1 km away**
- Tracks multiple drones at once, full 360° coverage
- Automatically zooms in and follows the drone's path

Anduril Roadrunner-M (USA)

- A small jet-powered drone that **hunts and destroys other drones**
- Launches in seconds from a box called the “Nest”
- Flies at high speed, locks onto the target, destroys it with a warhead
- If it turns out to be a false alarm — it flies back and lands safely
- Cost: ~\$500,000 per unit

Anduril Lattice AI (The Brain)

- Software that connects all sensors (cameras, radar, satellites) into one screen
- AI automatically detects, classifies, and tracks threats
- One operator can manage the whole system from a laptop

Key takeaway for our project: Real systems use multiple sensors together. Our Raspberry Pi version focuses on the **camera + AI** layer of this.

3. The Camera — Raspberry Pi Camera Module 3 NoIR

Feature	Spec
Sensor	Sony IMX708, 12MP
Resolution	Up to 4608×2592
Video	1080p @ 50 FPS, 720p @ 100 FPS
Field of View	75° (standard) or 120° (wide)
Night Vision	Yes (NoIR = no IR filter, works with IR LEDs)
HDR	Yes
Price	~\$25

Detection Range: Realistically detects a drone clearly up to **100–150 metres** with standard lens. For longer range, add a USB telephoto camera.

For night use: pair the NoIR camera with **850nm IR LED floodlights** — invisible to humans, visible to the camera.

4. The AI Model — YOLOv8 (You Only Look Once)

Why YOLO? It's fast, accurate, and runs on small hardware like the Raspberry Pi.

Which version for RPi 4? → YOLOv8n (nano) — smallest and fastest version, optimized for limited hardware.

What it does: Looks at each camera frame, draws a box around the drone, and says “drone detected — 94% confidence.”

Performance target: - mAP50 (accuracy score): **above 0.90** (90%+) - FPS on RPi 4: **5–15 FPS** (stable, usable for real-time detection) - To boost FPS: convert model to **INT8 format** using TFLite or ONNX

5. The Dataset — What to Train On

We need images of drones AND non-drones so the model learns the difference.

Classes to include:

Class	Why
Drone / UAV	Main target to detect
Bird	Looks similar on camera, must not trigger false alarm
Airplane	Different flight pattern, must be ignored
Helicopter	Rotating blades — similar to drone but bigger
Balloon	Slow-moving, non-threat

Where to get the data (free):

- **Roboflow Universe** → search “drone detection” → 2,600–22,000+ labelled images available
- **AOD-4 Dataset** → 22,516 images, 4 classes: drones, birds, helicopters, airplanes
- **Anti-UAV Dataset** → military-focused, infrared + visible light footage

Data augmentation (making dataset bigger/stronger):

Flip, rotate, blur, crop, and change brightness on existing images → this multiplies your dataset size and makes the model more robust.

6. Training the Model

Step	What to Do
1. Collect dataset	Download from Roboflow (5,000–10,000 images minimum)
2. Annotate	Label boxes around drones in every image (Roboflow auto-annotates)
3. Train	Use Google Colab (free GPU) — train for 150 epochs
4. Export	Export to TFLite or ONNX format for Raspberry Pi
5. Deploy	Copy model to RPi, run with Python + OpenCV

Target accuracy: mAP50 > 0.90 achieved consistently with 150 epochs + augmentation.

7. How False Detections Are Avoided

The biggest challenge: **drones and birds look almost identical on radar and camera.**

Object	How We Tell It Apart From a Drone
Bird	Irregular flapping movement, no rigid frame shape, different flight pattern
Airplane	Much faster, fixed wings, flies in straight lines at high altitude

Object	How We Tell It Apart From a Drone
Balloon	Drifts with wind, no motor, no directional flight
Helicopter	Larger, slower rotation, very different size profile

Techniques used: - Multi-class training (model sees all these objects during training) - Confidence threshold set to **>0.75** (ignores weak detections) - Behavioral tracking: if object moves like a drone (directional, fast, hovering) → alert

8. Full System Flow (How It All Connects)

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Camera (Pi Cam v3 NoIR)
  ↓
Raspberry Pi 4 (4GB)
  ↓
YOLOv8n model processes each frame
  ↓
Drone detected? → Trigger alert / log / send coordinates
  ↓
Human operator reviews → Authorizes response
  ↓
Counter-drone action (e.g., Roadrunner-M intercept or jammer)

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9. Budget (Basic Setup)

Item	Estimated Cost
Raspberry Pi 4 (4GB)	~\$55
Camera Module 3 NoIR	~\$25
IR LED floodlight (850nm)	~\$15
MicroSD card (64GB)	~\$10
Power supply + case	~\$15
Google Coral USB TPU (optional, boosts FPS)	~\$60
Total	~\$120–\$180

10. Key Takeaways

- Real military systems (AeroEye, Anduril) use **multiple sensors** — radar, RF, cameras, acoustics — fused together. Our system covers the **visual/AI layer**.
- **YOLOv8n** is the right model for Raspberry Pi 4 — fast, accurate, lightweight.
- Train on **5,000–10,000 images** with drone, bird, airplane, balloon, helicopter classes.
- **150 epochs + data augmentation** consistently hits 90%+ accuracy.
- The **Camera Module 3 NoIR** covers day and night detection up to ~150m.
- Export to **TFLite INT8** to keep FPS stable on the Pi.
- Always run a **confidence threshold of 0.75+** to kill false positives.

Report prepared for internal military internship research purposes.